Synthesis Review: Taylor Polynomials and Remainder Bounds

Building Taylor polynomials, reusing known series, and bounding the remainder

Directions

- Problems 1–3 are a warm-up, one on each idea. After that the three are mixed — decide first whether you should differentiate, substitute into a known series, or bound an error.
- Building from scratch: $P_n(x) = \sum_{k=0}^{n} \dfrac{f^{(k)}(a)}{k!}(x-a)^k$. Reusing a known series is almost always faster when one exists.
- Two remainder bounds. The Lagrange form $|R_n| \leq \dfrac{M|x-a|^{n+1}}{(n+1)!}$ always applies, with M a bound on the next derivative over the whole interval. The alternating bound — error at most the first omitted term — applies only when the series alternates.
- Round decimals to five places unless told otherwise.

1. Let $f(x) = e^{2x}$, with third-degree Maclaurin polynomial $P_3(x) = 1 + 2x + 2x^2 + \frac{4}{3}x^3$.

(a) Find $f'''(x)$, the derivative that produces the cubic coefficient.

(b) Use P_3 to estimate $f(0.1)$.

Answer: ________________

2. Using the Maclaurin series for $\cos x$, the series for $x \cos x$ begins $x - \frac{x^3}{2} + \frac{x^5}{24}$. Use those three terms to estimate the value of $x \cos x$ at $x = 0.4$.

Answer: ________________

3. The third-degree Maclaurin polynomial for e^x is used to estimate e^x on the interval $0 \leq x \leq 1$.

(a) Compute the Lagrange error bound at $x = 1$, taking $M = e$.

(b) Explain why $M = e$ is the right bound to use on this interval.

Answer: ________________

4. Let $f(x) = \ln x$, expanded about $a = 1$, so $P_3(x) = (x-1) - \dfrac{(x-1)^2}{2} + \dfrac{(x-1)^3}{3}$.

(a) Find $f'''(x)$.

(b) Use P_3 to estimate $\ln(1.2)$.

Answer: ________________

5. Substituting into the geometric series $\dfrac{1}{1-u} = 1 + u + u^2 + \cdots$ with $u = -3x^2$ gives $\dfrac{1}{1+3x^2} = 1 - 3x^2 + 9x^4 - \cdots$. Use those three terms to estimate the value at $x = 0.2$.

Answer: ________________

6. The Maclaurin series for $\cos x$ is used through the x^4 term to estimate $\cos(0.5)$.

(a) Compute the error bound the alternating series remainder guarantees.

(b) Explain what you had to check about the series before using that bound.

Answer: ________________

7. Let $f(x) = \sqrt{x}$, expanded about $a = 4$, so $P_2(x) = 2 + \dfrac{x-4}{4} - \dfrac{(x-4)^2}{64}$.

(a) Find $f''(x)$, the derivative that produces the quadratic coefficient.

(b) Use P_2 to estimate $\sqrt{4.4}$.

Answer: ________________

8. Substituting $x = -1$ into the Maclaurin series for e^x gives $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$.

(a) Write the exact sum of that series, and a decimal to four places.

(b) If the first four terms ($n = 0$ through $n = 3$) are used as an estimate, give the exact error bound the alternating series remainder guarantees.

Answer: ________________

9. The third-degree Maclaurin polynomial for $f(x) = \ln(1 + x)$ is used to estimate $f(x)$ on $0 \le x \le 0.5$. Its fourth derivative is $f^{(4)}(x) = -\frac{6}{(1+x)^4}$.

(a) Compute the Lagrange error bound at $x = 0.5$.

(b) Explain how you chose M, and why the largest value of $|f^{(4)}|$ occurs where it does.

Answer: ________________

10. Let $f(x) = xe^x$, with third-degree Maclaurin polynomial $P_3(x) = x + x^2 + \dfrac{x^3}{2}$.

(a) Find $f'''(x)$, and check that it gives the cubic coefficient shown.

(b) Use P_3 to estimate $f(0.3)$.

(c) Which remainder bound — the Lagrange form or the alternating-series form — would you use to measure the error in part (b), and why is the other one unavailable here?

Answer: __________________